A Survey on AI and Machine Learning Techniques for Phishing Detection, Malware Analysis, and Human-Centered Cybersecurity: Toward a Unified Real-Time Monitoring Dashboard

Sreedevi R P, Vishnu V Nair, Abhi Raj
Varsha C Mohan, Geetha S, Suvarna Dev
Department of Computer Engineering
College of Engineering Chengannur
APJ Abdul Kalam Technological University, Kerala, India

Abstract*— Modern organizations face a rapidly evolving cyber threat landscape in which phishing emails and malicious URLs, polymorphic and fileless malware, and increasingly convincing AI-generated deception coexist with persistent human and organizational weaknesses. Detection research has historically been fragmented across narrow sub-domains such as URL classification, malware image analysis, and security-awareness training, with few systems combining technical detection, economic justification, and human-centered explanation into a single operational tool. This survey examines recent research (2024–2026) spanning four themes: machine-learning and large language model (LLM) based phishing and URL detection, static/dynamic/image-based malware analysis, human-centered and organizational cybersecurity, and the economics of security investment. The reviewed works are compared on their core approach, models used, datasets, and reported performance. The analysis shows that while individual detection techniques have matured and now report accuracies in the 90–99% range, most remain single-purpose, evaluated offline on static datasets, and disconnected from the real-time alerting and human-awareness layer that determines whether a detection actually leads to protective action. This gap indicates the need for a web-based cybersecurity monitoring dashboard that combines lightweight machine-learning and LLM-assisted threat classification with real-time alerting, so that detection accuracy can be translated into timely, explainable, and actionable security awareness for end users and small teams. Such a system is proposed as a direction for future work at the end of this survey.

***Keywords*—** *Cybersecurity, Phishing Detection, Malware Detection, Machine Learning, Large Language Models, Threat Intelligence, Security Awareness, Real-Time Monitoring.*

I. INTRODUCTION

Cyberattacks against individuals, software development organizations, and financial institutions continue to grow in both volume and sophistication. Phishing remains the dominant initial-access vector, malware authors increasingly rely on obfuscation, fileless in-memory execution, and polymorphism to evade signature-based tools, and the emergence of generative AI has introduced a new class of highly realistic synthetic threats such as deepfakes and AI-written phishing content. At the same time, technical detection accuracy alone does not guarantee protection: security is also a human and organizational problem, shaped by user awareness, training, and the economic case an organization can make for investing in defenses.

Machine learning and deep learning have become the dominant tools for automating phishing URL and email classification, and for detecting malware from static, dynamic, and image-based representations of executables. More recently, large language models (LLMs) have been applied not only to classify threats but to generate human-readable explanations for why a message or file was flagged, directly addressing the trust and interpretability gap that purely statistical classifiers leave open. In parallel, researchers have begun to study the human side of the problem through structured literature reviews of organizational challenges, immersive virtual-reality training environments, and economic models that quantify the return on cybersecurity investment.

This survey reviews recent literature published between 2024 and 2026 spanning three themes: (1) machine-learning and LLM-based phishing and malicious-URL detection, (2) static, dynamic, and image-based malware detection and forensic analysis, and (3) human-centered, organizational, and economic dimensions of cybersecurity. For each paper, the approach, the models or methods used, and the reported results are summarized, followed by a comparative analysis and a discussion of the gaps that remain unaddressed. Based on these identified gaps, the survey concludes by proposing the outline of a system capable of addressing them.

II. REVIEW OF RECENT PHISHING, MALWARE, AND HUMAN-CENTERED CYBERSECURITY RESEARCH

A. Machine Learning and LLM-Based Phishing and URL Detection

Misiek and Hyla evaluated an XGBoost-based phishing URL detection method across three independent datasets, a custom set of 75,738 samples, the large-scale GramBeddings

dataset of 639,723 samples, and the PhiUSIIL dataset of 47,103 samples, to test cross-dataset generalization rather than single-dataset accuracy [3]. The gradient-boosted model reached 90.7% accuracy and a 91.2% F1-score on the custom dataset, with a lower 77.8% average accuracy across all three test sets, highlighting how feature instability degrades performance when a model trained on one distribution is applied to another. Compared against a neural network baseline, XGBoost achieved 2.1% higher accuracy while training 12.6 times faster (0.95s vs. 12.01s) and using 3.6 times less memory, although the neural network offered faster per-sample inference. The main limitation is that cross-dataset accuracy still drops substantially relative to in-distribution accuracy, showing that feature engineering choices must be revisited whenever a detector is deployed on traffic unlike its training data.

Koide, Nakano, and Chiba proposed ChatSpamDetector, an LLM-based phishing email detection system that analyzes email headers and body text to produce both a classification and a human-readable rationale for its decision [4]. Evaluated against multiple LLMs and baseline approaches, the system achieved 99.70% accuracy, and the authors found that prompt design substantially affected detection performance. By generating and analyzing natural-language rationales, the system directly targets the interpretability gap left by purely statistical filters, but it inherits the latency and API-cost limitations common to LLM-based pipelines and depends heavily on careful prompt engineering to sustain its reported accuracy.

Aldhuwayhi and Alromih proposed a dynamic phishing URL detection framework that combines a machine-learning classifier with a crowdsourced feedback mechanism, structured around feature extraction from URLs and user reports, ML-based classification, and continuous model refinement from crowdsourced signals [6]. The system reached 99.8% accuracy on a recent phishing URL dataset and was shown to outperform static detection baselines evaluated under the same conditions. The crowdsourcing component allows the model to adapt to newly observed attack patterns without full retraining, but it introduces dependence on the volume and reliability of user-submitted reports, and the paper does not evaluate robustness against adversarial or malicious crowd input.

Duarte et al. developed a machine-learning framework for early detection of phishing URLs hosted on newly registered and parked domains, applied in collaboration with a financial institution and trained on a curated dataset of 211,659 URLs drawn from real-time SSL certificate monitoring, domain listings, and phishing incident reports [11]. A Light Gradient Boosting Machine classifier achieved 96.02% recall and 97.28% accuracy on a balanced dataset under 10-fold cross-validation, and feature selection reduced model complexity while preserving detection effectiveness, supporting practical deployment. This work is notable for targeting the earliest possible detection point, newly registered and parked domains, before a phishing campaign is fully weaponized, but its evaluation is limited to one financial institution's traffic and data-collection pipeline.

B. Static, Dynamic, and Image-Based Malware Detection

Elgarduh et al. proposed Hybrid Feature Image Mapping (HFIM), a multi-channel malware visualization technique that encodes structural (binary content), semantic (opcode transition), and behavioral (API activity) information into a single RGB image so that convolutional neural networks can learn cross-domain patterns [2]. Evaluated against three representative malware-imaging baselines (HIT4Mal, MC-ISA, and MTV) across seven CNN architectures under 10-fold cross-validation, HFIM produced consistent improvements in accuracy, precision, recall, and F1-score, and remained effective when individual feature streams were partially unavailable. The main limitation is that image-based encoding still depends on successful static and dynamic feature extraction, so heavily obfuscated or packed samples that resist both analysis stages can weaken all three input channels simultaneously.

Gutierrez et al. proposed a hybrid deep-learning model for real-time automatic malware detection that combines static code features with dynamic runtime behavior, using convolutional neural networks for visual/static analysis and recurrent neural networks for sequential/dynamic analysis [7]. The combined model achieved 98% precision, 97% recall, and an F1-score of 0.975, outperforming traditional signature and heuristic methods (88–89% precision) and recent deep-learning approaches reporting up to 96% precision. While the reported results are strong, the paper evaluates the model on curated datasets rather than live production traffic, leaving open how well the CNN-RNN pipeline sustains real-time throughput under operational load.

Kara and Yılmaz proposed HAP-Analyzer, a hybrid forensic analysis framework for fileless malware that structurally integrates static, dynamic, and memory-based analysis rather than introducing a new detection algorithm [8]. Static analysis inspects binary properties and embedded components, dynamic analysis monitors runtime behavior and execution paths, and memory analysis recovers transient artifacts that exist only in volatile memory and are otherwise inaccessible; the findings from each stage inform the next, producing a cohesive, iterative investigation workflow. Demonstrated on a real-world fileless malware case study and evaluated using artifact-oriented metrics such as analysis duration and artifact-recovery effectiveness, HAP-Analyzer prioritizes interpretability, methodological transparency, and reproducibility over full automation, making it well suited to incident response and forensic investigation rather than real-time blocking.

Alshoulie and Mehmood conducted a comprehensive survey of deep-learning-based malware detection, synthesizing 109 publications from 2011 to 2024 and analyzing CNN, RNN, and hybrid CNN-RNN architectures across public datasets including BIG2015, Malimg, Drebin, Malicia, and

IoT-23 [10]. Reported detection accuracies across the surveyed literature range from 93% to 98.7%, with false-positive rates as low as 1.1% in the best hybrid models, and the survey contributes a structured taxonomy of architectures and an ablation-style summary of CNN versus RNN contributions. The survey's own stated limitations, adversarial robustness and computational complexity, echo across nearly all of the primary studies reviewed in this section, indicating that these remain open problems rather than solved ones.

C. Human-Centered, Organizational, and Economic Cybersecurity

Almarwani, Almarwani, and Almarwani introduced the AI-Cyber-User Awareness Framework, a socio-technical model grounded in Technology Threat Avoidance Theory (TTAT) that explains how AI-generated threats such as deepfakes and synthetic text interact with user cognition and protective behavior [1]. Built from a structured analysis of twenty-five peer-reviewed studies and validated with survey data from 144 participants in a Middle Eastern context, the study found a pronounced confidence-competence gap: mean objective detection accuracy for AI-generated deception was only 42.53%, despite moderate perceived preparedness, and detection competence emerged as the strongest predictor of protective cybersecurity behavior. This result is a direct caution against relying on awareness campaigns alone; the paper argues that measurable detection-skill training, not just risk messaging, is what changes behavior, but the study is correlational, survey-based, and geographically limited to one region.

Khattak et al. developed the Cyber Security Challenges in Software Development (CSC-SD) strategic model through a systematic literature review of 77 research articles drawn from IEEE, ACM, Springer, Google Scholar, and ScienceDirect [5]. The review identified eleven potential challenges facing software development organizations, including cyber-attack exposure, knowledge deficiency, framework gaps, technical support limitations, disaster-preparedness gaps, resource constraints, confidentiality and trust deficiency, management and measurement gaps, unauthorized-access risk, development-execution gaps, and quality/reliability concerns. As a literature-review-based model, CSC-SD is not empirically validated against real organizational outcomes, but it provides a structured checklist of organizational weak points that a monitoring tool could help surface and track over time.

He, Xin, and Luo proposed FAIR-ROSI, a cybersecurity investment model that integrates five qualitative and quantitative security metrics with the Factor Analysis of Information Risk (FAIR) framework to extend the traditional Return on Security Investment (ROSI) calculation [9]. Evaluated through a case study comparing pre- and post-control loss measures, FAIR-ROSI produced a narrower margin between actual and projected loss values and a significantly higher return on investment relative to total security expenditure than the traditional ROSI model alone. The model gives organizations a more granular, evidence-based way to justify security spending, but it depends on the availability of reliable loss and control-effectiveness data, which many smaller organizations do not systematically collect.

Abril et al. studied the use of virtual reality (VR) to train individuals to recognize and resist cyberattacks, exposing adult participants to a simulated corporate workday containing social-engineering and phishing scenarios, with control, feedback, and no-feedback experimental groups [12]. Participants who received performance feedback within the VR simulation showed a better understanding of cyber-threat complexity, and risky behaviors observed during the VR tasks, along with participant age, predicted attitudes toward cybersecurity and cybercrime in organizational settings. The findings support experiential, feedback-driven training as a complement to detection technology, though VR-based training requires specialized hardware and content development that limits its scalability compared to a lightweight, browser-based dashboard.

III. COMPARATIVE ANALYSIS

The reviewed works fall into three broad categories: machine-learning and LLM-based phishing and URL detection, static/dynamic/image-based malware detection and forensics, and human-centered or organizational-economic approaches to cybersecurity. Among the phishing-detection papers, classical gradient-boosted models such as XGBoost remain highly efficient and competitive in-distribution but lose substantial accuracy across datasets, while LLM-based systems such as ChatSpamDetector and crowdsourced ML pipelines push accuracy above 99% at the cost of added latency, API expense, or dependence on user-submitted signals. Among the malware-detection papers, image-based and hybrid CNN-RNN architectures consistently outperform single-domain static or dynamic analysis, and the broader survey by Alshoulie and Mehmood confirms this pattern across 109 independent studies, while HAP-Analyzer shows that forensic traceability, not just detection accuracy, is a distinct and underserved requirement for fileless threats. Finally, the human-centered and organizational papers converge on a consistent finding: technical detection accuracy does not automatically translate into protective behavior or justified investment, and closing that gap requires measurable skill training, structured organizational assessment, and economically grounded investment models. Table 1 summarizes the reviewed studies.

However, none of the reviewed systems combine automated phishing and malware detection, forensic traceability, human detection-competence measurement, and organizational investment justification within a single, real-time, low-cost monitoring platform. Therefore, a practical system can focus on integrating lightweight ML/LLM detection, explainable alerting, and human-centered feedback into one end-to-end cybersecurity monitoring pipeline.

TABLE 1
COMPARISON OF THE REVIEWED PHISHING, MALWARE, AND HUMAN-CENTERED CYBERSECURITY STUDIES

Ref.	Main Approach	Main Technology	Data/Platform	Key Result
[1]	Socio-technical framework for AI-deception awareness and detection competence.	TTAT-based survey model; correlation/regression/mediation analysis.	25-study literature synthesis; n=144 survey (Middle East).	Mean detection accuracy 42.53%; detection competence strongest predictor of protective behavior.
[2]	Multi-channel image-based malware visualization (static+dynamic fusion).	RGB image encoding + 7 CNN architectures.	10-fold CV vs. HIT4Mal, MC-ISA, MTV baselines.	Consistent gains in accuracy, precision, recall, F1 over single-domain baselines.
[3]	Gradient-boosted phishing URL classification with cross-dataset validation.	XGBoost vs. Neural Network baseline.	Custom (75.7K), GramBeddings (639.7K), PhiUSIIL (47.1K).	90.7% acc./91.2% F1 (custom); 77.8% avg. cross-dataset; 12.6× faster training than NN.
[4]	LLM-based phishing email detection with generated rationale.	LLM prompting (multi-model comparison).	Phishing/legitimate email datasets.	99.70% accuracy; prompt design strongly affects performance.
[5]	Systematic literature review identifying organizational cyber-risk challenges.	SLR of 77 articles across 6 databases.	Software Development Organizations (SDO) context.	11 challenge categories identified (e.g. knowledge, framework, resource gaps).
[6]	Dynamic phishing detection combining ML classification with crowdsourcing.	ML classifier + crowdsourced feedback loop.	Recent phishing URL dataset.	99.8% accuracy; outperforms static baselines under same conditions.
[7]	Hybrid static+dynamic malware detection.	CNN (visual/static) + RNN (sequential/dynamic).	Curated malware behavior dataset.	98% precision, 97% recall, F1 0.975; beats traditional (88–89%) methods.
[8]	Hybrid static, dynamic, and memory forensic analysis for fileless malware.	Structured multi-stage forensic correlation.	Real-world fileless malware case study.	Improved artifact recovery and forensic traceability vs. existing hybrid frameworks.
[9]	Cybersecurity investment model integrating FAIR risk metrics with ROSI.	Qualitative + quantitative FAIR-ROSI scoring.	Organizational case study (pre/post controls).	Narrower loss-projection margin and higher ROI than traditional ROSI.
[10]	Comprehensive survey of deep-learning malware detection literature.	CNN, RNN, hybrid CNN-RNN taxonomy.	109 publications (2011–2024); BIG2015, Malimg, Drebin, Malicia, IoT-23.	Accuracy range 93–98.7%; false-positive rate as low as 1.1% (hybrid models).
[11]	Early phishing-URL detection on newly registered/parked domains.	LightGBM classifier + feature selection.	211,659 URLs (SSL monitoring, domain lists, incident reports); financial institution.	96.02% recall, 97.28% accuracy (10-fold CV).
[12]	Immersive VR training for cybersecurity risk-behavior awareness.	VR simulation with feedback/no-feedback/control groups.	Adult participant study (higher-education sample).	Feedback improved threat comprehension; age and risky VR behavior predicted real-world attitudes.

IV. RESEARCH GAPS AND FUTURE DIRECTIONS

The reviewed studies show substantial progress in phishing and URL detection, malware detection and forensics, and human-centered or organizational cybersecurity considered individually. However, several gaps remain when these techniques are considered for a lightweight, real-time monitoring dashboard aimed at end users and small teams.

1) Fragmentation between detection and alerting:

Most phishing and malware detection papers reviewed report strong offline accuracy (90–99.8%) but stop at classification, without addressing how a positive detection reaches a human in time to act on it. A practical system needs a real-time alerting layer, such as webhook-based notifications, tightly coupled to the detection model rather than a standalone research classifier.

2) Trade-off between detection accuracy and explainability:

ChatSpamDetector and the AI-Cyber-User Awareness Framework both show that raw detection accuracy does not by itself build user trust or improve human detection competence; explanations and skill-building matter as much as classification correctness. A future system should generate a concise, human-readable rationale alongside every alert rather than a bare risk score.

3) Cross-dataset and cross-domain generalization is unresolved:

The XGBoost phishing study shows accuracy dropping from 90.7% in-distribution to 77.8% across datasets, and the malware-detection survey highlights adversarial robustness as an open problem across 109 studies. A deployed monitoring tool must be evaluated, and where possible retrained or recalibrated, on traffic resembling its actual users rather than relying solely on published benchmark numbers.

4) Static and dynamic analysis alone are insufficient for fileless and evasive threats:

HAP-Analyzer and HFIM both show that combining static, dynamic, and (for fileless malware) memory-based analysis outperforms any single analysis mode, but such integration is analyst-driven and not yet packaged for lightweight, automated deployment.

5) Human and organizational factors remain decoupled from technical tooling:

The CSC-SD model, the VR training study, and the AI-Cyber-User Awareness Framework each show that organizational challenges, detection competence, and behavior are measurable, but none of the reviewed technical detection systems incorporate these human-centered signals. A future system should track and surface human-facing metrics, such as how often users act correctly on alerts, alongside technical detection metrics.

6) Economic justification is treated separately from detection capability:

FAIR-ROSI demonstrates that investment can be justified with granular, evidence-based metrics, but this economic layer is not connected to any of the reviewed detection systems' own performance data, leaving organizations to build the business case for a tool manually.

7) Lack of a single, lightweight, integrated platform:

The major gap identified across the reviewed studies is the absence of one accessible architecture that combines ML/LLM-based phishing and malware detection, real-time explainable alerting, and human-centered feedback for individuals and small teams, rather than large enterprises with dedicated security operations centers. Future work can focus on building such a platform.

V. CONCLUSION

This survey reviewed recent literature covering machine-learning and LLM-based phishing and URL detection, static/dynamic/image-based malware detection and forensics, and human-centered, organizational, and economic dimensions of cybersecurity. These works show that gradient-boosted and deep-learning models achieve high in-distribution accuracy for phishing and malware classification, that combining static, dynamic, and memory-based analysis improves both detection and forensic traceability, and that technical accuracy alone does not guarantee protective behavior or justified investment without explanation, training, and economic modeling. However, none of the reviewed systems combine these capabilities into a single, real-time, explainable monitoring tool accessible to individuals and small teams.

These identified gaps, fragmented detection and alerting, a trade-off between accuracy and explainability, unresolved cross-dataset generalization, insufficient handling of fileless and evasive threats, human and organizational factors decoupled from technical tooling, and economic justification treated separately from detection, together motivate a system that can address all of them at once. Such a system can be envisioned as a web-based cybersecurity monitoring dashboard that pairs lightweight machine-learning and LLM-assisted threat classification with real-time webhook-based alerting, aiming to turn detection accuracy into timely, understandable, and actionable notifications. By combining efficient detection, human-readable explanation, and real-time delivery, this proposed system would bring together approaches validated individually in the reviewed literature into a unified, accessible platform for this domain.

REFERENCES

[1] R. Almarwani, M. Almarwani, and F. Almarwani, "AI-Driven Synthetic Threats in Cybersecurity: A User-Centered Framework for Awareness, Detection, and Protective Behavior," *IEEE Access*, vol. 14, 2026, doi: 10.1109/ACCESS.2026.3673980.

[2] A. Elgarduh, A. Zainal, F. A. Ghaleb, S. N. Qasem, A. M. Albarrak, and F. Saeed, "Hybrid Feature Image Mapping (HFIM): A Multi-Channel Image Approach for Malware Detection Using Static and Dynamic Features," *IEEE Access*, 2026 (in press).

[3] M. Misiek and T. Hyla, "XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation," *IEEE Access*, vol. 14, 2026, doi: 10.1109/ACCESS.2026.3672690.

[4] T. Koide, H. Nakano, and D. Chiba, "ChatSpamDetector: Large Language Model-Based Phishing Email Detection," *IEEE Access*, vol. 14, 2026, doi: 10.1109/ACCESS.2026.3704566.

[5] K. Khattak, Sabahat, N. Iqbal, Zeeshan, and J. Choi, "Cyber Security Challenges: A Strategic Model for the Software Development Organizations," *IEEE Access*, vol. 14, 2026, doi: 10.1109/ACCESS.2026.3695471.

[6] L. Aldhuwayhi and A. Alromih, "A Dynamic Phishing URL Detection System Integrating Advanced Machine Learning and Crowdsourced Threat Intelligence," *IEEE Access*, 2026, doi: 10.1109/ACCESS.2026.3700833.

[7] R. Gutierrez, W. Villegas-Ch., L. Naranjo Godoy, A. Mera-Navarrete, and S. Luján-Mora, "Application of Deep Learning Models for Real-Time Automatic Malware Detection," *IEEE Access*, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3436588.

[8] I. Kara and K. Yılmaz, "HAP-Analyzer: A Hybrid Static, Dynamic, and Memory Analysis Approach for Fileless Malware," *IEEE Access*, vol. 14, 2026, doi: 10.1109/ACCESS.2026.3659346.

[9] Y. He, T. Xin, and C. Luo, "Enhancing Cybersecurity Investment with FAIR-ROSI: A Responsible Cybersecurity Approach to Digital Society," *Information Systems Frontiers*, vol. 28, pp. 697–712, 2026.

[10] M. Alshoulie and A. Mehmood, "Deep Learning Approaches for Malware Detection: A Comprehensive Review of Techniques, Challenges, and Future Directions," *IEEE Access*, vol. 13, 2025, doi: 10.1109/ACCESS.2025.3582875.

[11] J. D. Duarte, P. Chagas Junior, J. P. Javidi da Costa, E. J. da Costa, L. Peotta de Melo, R. Rabelo Nunes, C. G. V. N. Soares, and T. E. da Cunha Silva, "Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution," *IEEE Access*, vol. 13, 2025, doi: 10.1109/ACCESS.2025.3599454.

[12] T. Abril, P. Gamito, C. da Motta, J. Oliveira, F. Dias, F. Pinto, et al., "Exploring a Novel Approach to Cybersecurity: The Role of Ecological Simulations on Cybersecurity Risk Behaviors," *Virtual Reality*, vol. 29, art. 150, 2025.